



Lecture

13

Estimation Models-EOQ



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- Stock of items kept to meet future demand.
- Purpose of inventory management
 - How many units to order
 - When to order
- **Example:-** A shop keeps **50 mobile phones** in the store. These phones are kept so customers can buy them anytime. Those 50 phones are the **inventory** of the shop.



Two Forms of Demand

- Dependent
 - Demand for items used to **produce final products**.
 - Tires for autos are a dependent demand item.
- Independent
 - Demand for items used by external customers.
 - Cars, appliances, computers, and houses are examples of independent demand inventory.

- Carrying cost or Holding Cost
 - cost of holding an item in inventory.
- Ordering cost
 - cost of **replenishing inventory**.
- Shortage cost
 - temporary or permanent loss of sales when demand cannot be met.

- **Continuous system (fixed-order-quantity):-**

- Constant amount ordered when inventory declines to predetermined level.

➤ **Example:-** When stock becomes 50 sugar packets, the system automatically orders more.

- **Periodic system (fixed-time-period):-**

- Order placed for variable amount after fixed passage of time.

Example:- Stock is checked **once every month**. After checking, the owner finds 20 sugar packets left. He orders enough sugar for the next month.



Economic Order Quantity (EOQ) model

- Estimation models in Engineering Economics are used to estimate costs and make optimal decisions. **One of the most important estimation models in inventory management is the Economic Order Quantity (EOQ) model.**
- **EOQ helps us decide:**
 - *How much quantity should be ordered each time so that the **total inventory cost is minimum.***
- **Economic Order Quantity (EOQ)** is the **optimal order size** that minimizes the **total cost of inventory**, which includes:
 - Ordering cost
 - Holding (carrying) cost
 - Purchasing cost



Multiperiod model – The Economic Order Quantity



- Demand is known and deterministic: D units/year
- We have a known ordering cost, S , and immediate replenishment
- Annual holding cost of average inventory is H per unit
- Purchasing cost C per unit

Assumptions of the EOQ Model

The EOQ model is based on the following assumptions:

- Demand is known, deterministic, and constant.
- Ordering cost per order is fixed.
- Holding cost per unit per year is constant.
- **Replenishment occurs instantly** with zero lead time.
- No shortages or stock-outs are allowed.
- Unit purchase cost remains constant.



What is the optimal quantity to order?

Cost Components in EOQ Model

$$\text{Total Cost} = \text{Purchasing Cost} + \text{Ordering Cost} + \text{Inventory Cost}$$

$$\text{Purchasing Cost} = (\text{total units}) \times (\text{cost per unit})$$

$$\text{Ordering Cost} = (\text{number of orders}) \times (\text{cost per order})$$

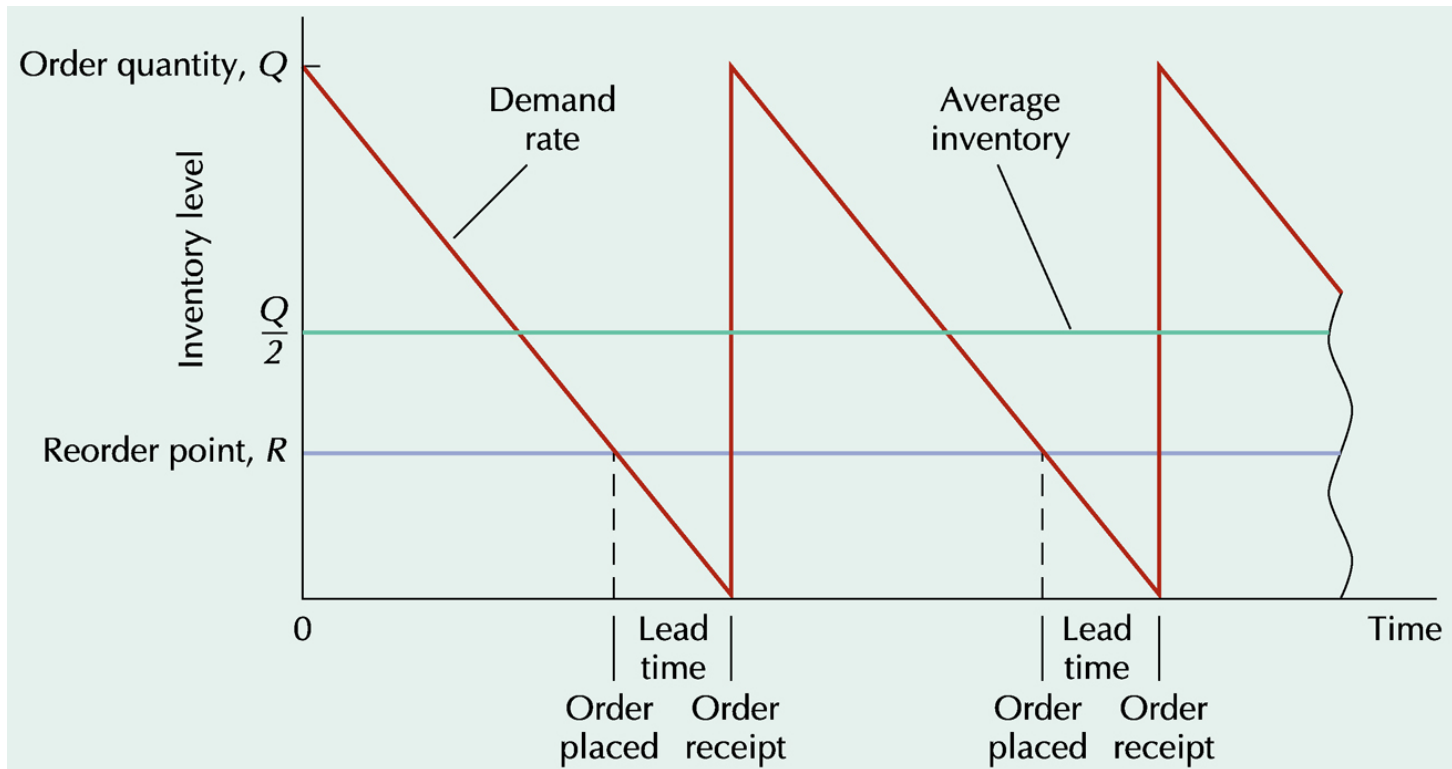
$$\text{Inventory Cost} = (\text{average inventory}) \times (\text{holding cost})$$



Finding the optimal quantity to order...

Let's say we decide to order in batches of Q ...

Inventory position



$$\text{Number of orders} = \frac{D}{Q} = \frac{\text{Total units needed}}{\text{Units per order}}$$

The average inventory for each period is... $\frac{Q}{2}$

This value is used to calculate **holding (carrying) cost**.

Lead time = time between **placing the order** and **receiving the order**.



Finding the optimal quantity to order...

- This is the cost of buying the required items to meet annual demand.

$$\text{Purchasing cost} = D \times C$$

- Each time an order is placed, a fixed ordering cost is incurred. If Q units are ordered each time, the number of orders per year is D/Q .

$$\text{Ordering cost} = \frac{D}{Q} \times S$$

- Holding cost is incurred for storing inventory. Since inventory decreases uniformly from Q to 0, the average inventory held is $Q/2$.

$$\text{Inventory cost} = \frac{Q}{2} \times H$$



So what is the total cost?

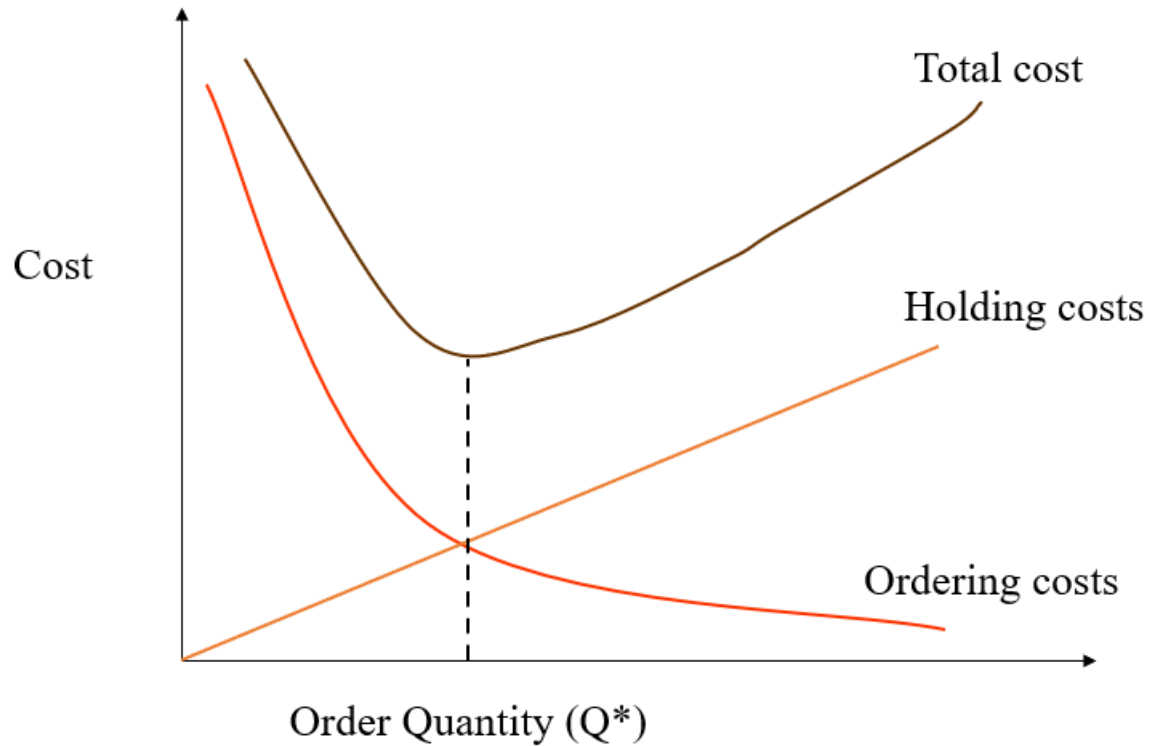
$$TC = DC + \frac{D}{Q} S + \frac{Q}{2} H$$

In order now to find the optimal quantity we need to **optimize the total cost with respect to the decision variable** (the variable we control).



What is the main insight from EOQ?

There is a tradeoff between holding costs and ordering costs



- ✓ As Q^* increases, the **number of orders per year decreases**, so ordering cost **reduces**.
- ✓ As Q^* increases, more inventory is stored, increasing the **holding cost**.
- ✓ Initially, total cost decreases because ordering cost falls rapidly.
- ✓ After a point, total cost increases **because holding cost dominates**.



Determination of Economic Order Quantity (Q^*)

To find the optimal order quantity, the total cost function is minimized with respect to Q . Differentiating TC and equating to zero gives:

$$Q^* = \sqrt{\frac{2SD}{H}}$$

Example:

Assume a car dealer that faces demand for 5,000 cars per year, and that it costs 15,000 to have the cars shipped to the dealership. Holding cost is estimated at 500 per car per year. How many times should the dealer order, and what should be the order size?

$$Q^* = \sqrt{\frac{2(15,000)(5,000)}{500}} = 548$$



Example

A company requires 16000 units of raw material costing Rs. 2 per unit. The cost of placing an order is Rs. 45 and the carrying cost is 10% per year per unit of the average inventory. Determine the economical order quantity and cycle time.



Example

$$EOQ = Q^* = \sqrt{\frac{2DS}{H}}$$

$$Q^* = \sqrt{\frac{2 \times 16000 \times 45}{0.20}}$$

$$Q^* = \sqrt{\frac{1,440,000}{0.20}}$$

$$Q^* = \sqrt{7,200,000}$$

$$Q^* = 2683 \text{ Units}$$

$$\text{No. of orders per year} = \frac{\text{Annual Consumption}}{EOQ}$$

$$\text{Number of orders} = \frac{D}{Q^*} = \frac{16000}{2683}$$

$$= 5.96 = 6 \text{ Orders/year}$$



Example

1. An auto parts supplier sells Hardy-brand batteries to car dealers and auto mechanics. The annual demand is approximately 1,200 batteries. The cost is Rs. 28 for each battery and the holding cost is 30 percent of the battery's cost. It costs approximately Rs. 20 to place an order (managerial and clerical costs). The supplier currently orders 100 batteries per month.
 - a. Determine inventory costs for the current order quantity.
 - b. Determine the economic order quantity (EOQ).
 - c. How many orders will be placed per year using the EOQ?



Sol.

Annual demand: $D = 1200$ batteries per year

Item cost: $c = \text{Rs. } 28$ per battery

Holding cost: $H = ic = 0.30(28) = \text{Rs. } 8.40$ per battery per year

Order cost: $S = \text{Rs. } 20$ per order

Current order quantity: $Q = 100$ batteries

a. Total cost = $(D/Q)*S + (Q/2)*H$ Ordering cost + Holding cost
 $= (1200*100)/20 + (100/2)* (8.40) = 240 + 420 = \text{Rs. } 660.$

b. $EOQ = \sqrt{2D \cdot S / H}$

$$EOQ = \sqrt{2 * 1200 * 20 / 8.40}$$

$$= 75.6 = 76 \text{ batteries.}$$

No. of orders = $D/EOQ = 1200/76 = 15.8$ orders per year = 16 orders